

emteqPRO: Face-mounted Mask for Emotion Recognition and Affective Computing

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ABSTRACT

The paper presents the emteqPROtm system, which uses a face-mounted multi-sensor mask to measure the facial physiological responses, facial muscle activations, and motions from the user. These responses are then analyzed by machine learning algorithms to recognize and better understand the user's affective state and context, i.e., emotions, arousal, valence, stress response, activities, etc. The system can work by itself, as an open mask, or can be combined with a commercial Virtual Reality head mounted display. It comprises 3 sensor modalities: a 7-contact f-EMG sensor, a PPG sensor, and a 3-axis IMU, enabling it to measure the affective state of the user in real time. We will demonstrate how the system is used in practice in a Virtual Reality environment. This newly developed technology has the potential to significantly improve the way we collect data, design experiences, and interact within Virtual, Mixed and Augmented Realities.

KEYWORDS

Machine learning, Sensors and actuators, Ubiquitous and mobile computing, Affective computing, Facial Expressions, Emotion Recognition, Arousal, Valence, EMG, PPG, IMU, Machine Learning

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1 EMOTION RECOGNITION

Emotions are subjective states of mind and are usually reactions to internal stimuli or events that occur around us. If strong enough, they are also expressed by our body, so other people can recognize them [1], e.g., we flush when we are embarrassed. However, if the emotions are at the milder end of the spectrum, they are difficult to detect and recognize because visually the body is not expressing symptoms, e.g., when we see a product on a shelf that catches our eye or when we are in a conversation.

Emotion recognition is one of the areas where computer systems still lag human performance, but there has been significant progress in the last decade — especially in the development and availability of wearable systems. The most traditional way of recognizing emotions is by using cameras [2]. These systems have inspired great interest, especially for situations where the emotions are expressed on the face. However, this approach cannot recognize subtle responses (e.g., increased muscle tension or heart rate changes when embarrassed). Furthermore, the theoretical basis of camera-based recognition of affective responses has been strongly discredited by Barrett et al [3]. They showed that unlike cameras, EMG is a scientifically valid method of evaluating affective states.

In recent years we have seen approaches that attempt to assess emotional responses using wearables that measure person's physiological signals (e.g., heart activity and sweating [4]). However, although these approaches can detect affective responses that are expressed by changes in heart activity or skin conductance (arousal), they cannot perform low latency detection of emotional responses that result in facial expression (i.e., changes in valence)."

To tackle these challenges, we present a novel multimodal approach and system - emteqPROtm (shown in Figure 1), which is the latest version of the Emteq's emotion analytics devices [5-6]. This system captures the person's heart activity, but also facial

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Figure 1: The emteqPRO mask mounted on a VR device (left), and a calibration scene showing the status of the sensors connectivity (right).



Figure 2: Left: The emteqPRO multi-sensor mask, showing the EMG channels: frontalis (in red), orbicularis (in green), zygomaticus (in blue) and corrugator (in purple). Right: The locations of the sensors mapped on the face.

expressions - and this way it provides more comprehensive information about the person’s emotional and affective state. Our proprietary Artificial Intelligence (AI) Emotion Engine combines this multimodal sensory data with advanced data fusion Machine Learning (ML) algorithms to recognize and better understand the user’s affective and emotional state.

2 EMTEQPRO SYSTEM

The emteqPRO system consists of a: (i) emteqPRO multi-sensor wearable mask (shown in Figure 2); (ii) emteqVR SDK; (iii) Super-Vision Application; (iv) AI Emotion Engine. The 4 main modules of the system are explained below.

The emteqPRO mask (shown in Figure 1 on the left) is a medical-grade wearable mask that detects the symptoms of emotional state through its integrated physiological sensors. It contains a photoplethysmographic (PPG) sensor, 7 contact facial electromyographic (f-EMG) sensors, and a 3-axis Inertial Measurement Unit (IMU) sensor. This way, it provides data on facial muscle activation at 7 locations on the user’s face (see Figure 1), pulse measurements and information on the head, and upper body motion. The mask may be integrated with commercial VR devices (the one shown in Figure 1 is for HTC Vive Pro) or may be used stand-alone in “Open Face” mode.

The emteqVR SDK - is a software development kit for the Unity 3D environment, providing integration between the immersive experience and the biometric data. It allows developers to access sensor data from the mask for real-time use and provides bi-directional communication with the SuperVision app.

The SuperVision - is an application for monitoring data and the progress of an immersive experience in real-time, providing in-flight supervision of emotional experiences and helping to ensure the highest quality of data recording. It allows researchers to monitor mask fit, data quality and the session in progress, and displays the contact state of sensors together with real-time sensor data.

The AI Emotion Engine - is proprietary Artificial Intelligence (AI) used to analyze the multimodal sensor data using advanced data-fusion and ML algorithms, and in doing so, to recognize the user’s affective and emotional state. In particular, there are several modules: (i) *heart-rate variability (HRV)* module – performs signal processing to extract a dozen HRV features; (ii) *arousal recognition ML model* – primarily uses the HRV features and data fusion for the IMU and EMG data; (ii) *valence recognition ML model* – similar to the arousal model, but more focused on the EMG data; (iv) *expression recognition ML model* – uses the EMG data to recognize facial expressions such as smile, frown and surprise.



Figure 3: The main menu of the VR scenes developed.

3 DEMO

The demo will show how the system is used in practice. For these purposes, we have developed a VR experience specifically designed to induce various emotional responses to users (Figure 3). First, we will show how the emteqPRO mask is placed on the face and how to check the mask's fit. Then, we will show the VR environment, where the user can choose different options. Next, we will show how the data is recorded and visualized in the SuperVision app. Finally, we will show the AI Emotion Engine that provides the insights about the user's affective state, i.e., arousal, valence, HRV analysis.

The system has various use cases, starting from researchers interested in affective computing. Then, architects can use our system to test various 3D designs and the affective response by the users. It can also be used by the gaming industry to measure the affective response to the games and the scenes by their users.

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REFERENCES

- [1] Shu, L.; Xie, J.; Yang, M.; Li, Z.; Li, Z.; Liao, D.; Xu, X.; Yang, X. A Review of Emotion Recognition Using Physiological Signals. *Sensors* 2018, 18, 2074.
- [2] J. Cohn and K. Schmidt, The Timing of Facial Motion in Posed and Spontaneous Smiles, *International Journal of Wavelets, Multiresolution and Information Processing*, 2(2): 121-132, 2004.
- [3] Barrett LF, Adolphs R, Marsella S, Martinez AM, Pollak SD. Emotional Expressions Reconsidered: Challenges to Inferring Emotion From Human Facial Movements. *Psychological Science in the Public Interest*. 2019;20(1):1-68.
- [4] C. A. Wang, T. Baird, J. Huang, J. D. Coutinho, D. C. Brien, and D. P. Munoz, "Arousal Effects on Pupil Size, Heart Rate, and Skin Conductance in an Emotional Face Task," *Frontiers in Neurology*, vol. 9, p. 1029, Dec. 2018, doi: 10.3389/fneur.2018.01029. Conference Name: ACM Woodstock conference
- [5] Mavridou I, McGhee JT, Hamed M, Fatoorechi M, Cleal A, Ballaguer-Balester E, Seiss E, Cox G, Nduka C. FACETE interface demo for emotion expression in VR. In 2017 IEEE Virtual Reality (VR) 2017 Mar 18 (pp. 441-442). IEEE.
- [6] Mavridou I, Seiss E, Hamed M, Balaguer-Balester E, Nduka C. Towards valence detection from EMG for Virtual Reality applications. in Proc. 12th Int. Conf. Disability, Virtual Reality and Associated Technologies (ICDVRAT 2018). Reading, U.K.: ICDVRAT, University of Reading, Sept. 2018, pp 1-4. ISBN:978-1-4503-0000-0/18/06

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